

Digital Research Toolkit for Linguists

Week 13: Managing large projects, literature research, AI and LLM for the humanities

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July 1, 2025

Psycholinguistics and Cognitive Modeling Lab

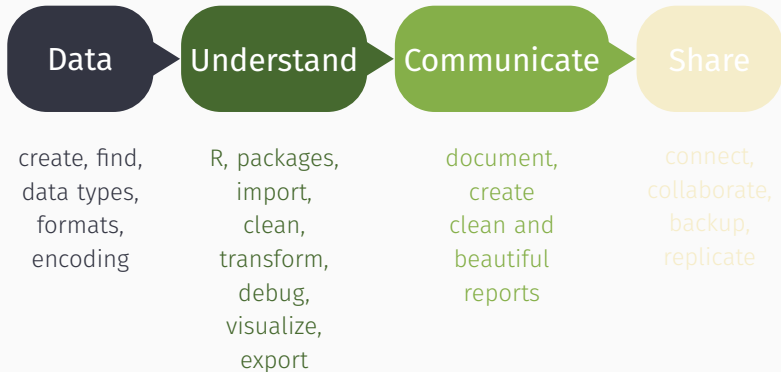


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5. AI and LLMs in Research
6. Homework assignment

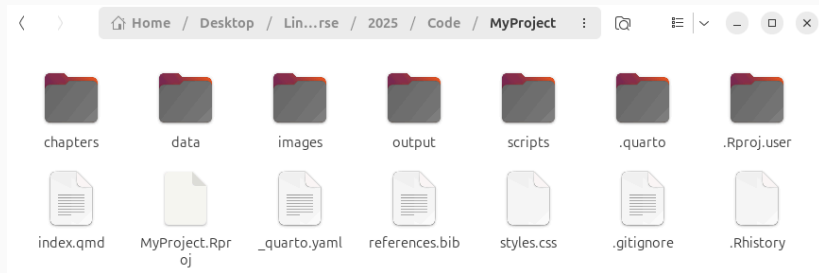
Large projects

New little big project

Writing everything in one file gets messy *fast*.

Solution: Split document into smaller files (e.g. chapters).

This is  especially relevant for books.



Offload all the preamble YAML stuff into `_quarto.yaml`

Your new project

New Project Wizard

Create Project



New Directory
Start a project in a brand new working directory >



Existing Directory
Associate a project with an existing working directory >

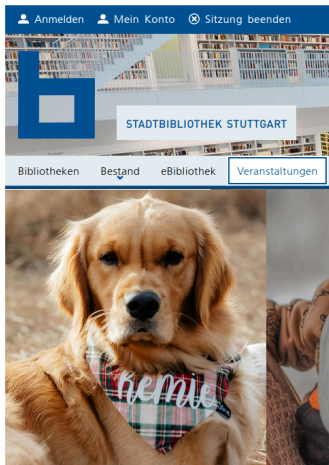


Version Control
Checkout a project from a version control repository >

Cancel

Bibliography

Bib basics: Not this



Reference files

Quarto uses Bib_{TEX} (or other programs) for creating and referencing bibliographies.

Workflow

Create bibliography → reference in text → generate references

Referencing and printing

In the preamble (YAML): `bibliography: references.bib`

The the main document (`index.qmd`):

```
# References
```

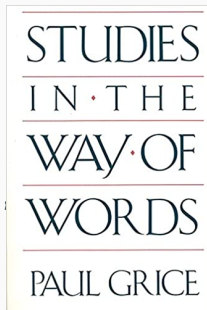
```
::: {#refs}
```

```
:::
```

Bib_{La}TeX bibliographies

Bib(La)TeX allows for using a separate `.bib` file with a list of references. Has a VERY rigid and defined structure

```
@book{grice1989,  
  title={Studies in the Way of Words},  
  author={Grice, Paul},  
  year={1989},  
  publisher={Harvard University Press},  
  address={Cambridge}  
}
```



Paul Grice (1989). *Studies in the Way of Words*. Cambridge: Harvard University Press

Referencing

Markdown	Output
Yadda yadda [see @grice1989, pp. 33-35; also @wickham2023]	Yadda yadda (see Grice 1989, 33–35; also Wickham, Çetinkaya-Rundel, and Grolemund 2023).
Blah blah [-@wickham2023]	Blah blah (2023)
In contrast, @grice1989	In contrast, Grice (1989)

Create from scratch

slow, tedious, but good for complicated entries

[http://tug.ctan.org/info/biblatex-cheatsheet/
biblatex-cheatsheet.pdf](http://tug.ctan.org/info/biblatex-cheatsheet/biblatex-cheatsheet.pdf)

Semi-automatic entry from template

many optional fields (e.g. translator, annotator, series) and others
you still need to fill in manually

Import online

fast, convenient, but **always** contain mistakes

Life's too short to write bibliography from scratch.

How Speakers Refer: The Role of Accessibility

Jennifer E. Arnold

First published: 31 March 2010 | <https://doi.org/10.1111/j.1749-818X.2010.00193.x> | Citations: 175

[Read the full text >](#)

PDF TOOLS SHARE

Abstract

One of the core components of language is reference. Speakers choose between expressions that are highly explicit (e.g., *Peter*) and reduced lexical forms (e.g., *he*). This paper reviews research driven by the *accessibility* or *salience* of the referent underlie these effects. Two classes of constraints are identified, which has traditionally been identified as the deictic constraints. Non-linguistic processing constraints that increase the use of explicit forms. These effects together support a modified version of the traditional claim that speakers choose referential explicitness so that the listener can identify the referent, and underscore the need for accessibility to be mediated by a non-linguistic representation.



Volume 4, Issue 4

April 2010

Pages 187-203

References



Related



Information

Recommended

Structured variation in child heritage speakers' grammars

Naomi Shin

Language and Linguistics Compass

Intransitivizing morphology

Raina Heaton

Language and Linguistics Compass

Description and analyses of nominal concord (Pt I)

Mark Norris

Doi-to-bib

DOI = (persistent) Digital Object Identifier

<https://www.doi2bib.org/>



doi2bib — give us a DOI
and we will do our best to get you the BibTeX entry

get BibTeX

```
@article{Arnold2010,  
  doi = {10.1111/j.1749-818X.2010.00193.x},  
  url = {https://doi.org/10.1111/j.1749-818X.2010.00193.x},  
  year = {2010},  
  month = mar,  
  publisher = {Wiley},  
  volume = {4},  
  number = {4},  
  pages = {187--203},  
  author = {Jennifer E. Arnold},  
  title = {How Speakers Refer: The Role of Accessibility},  
  journal = {Language and Linguistics Compass}  
}
```

<https://doi.org/10.1111/j.1749-818X.2010.00193.x>

Copy Bib to Clipboard

Copy URL to Clipboard

scholar.google.com

Artikel

Beliebige Zeit
 Seit 2022
 Seit 2021
 Seit 2018
 Zeitraum wählen...

Nach Relevanz sortieren

Nach Datum sortieren

Beliebige Sprache
 Seiten auf Deutsch

Alle Typen
 Übersichtsarbeiten

How speakers refer: The role of accessibility [\[PDF\] wiley.com](#)

[JE Arnold](#) - Language and Linguistics Compass, 2010, 4, S. 187-203.

One of the core components of language is referentiality. This paper reviews claims about the role of accessibility or saliency of the referent, and the effects of these on the choice of linguistic forms. Two classes of constraint are examined: (i) constraints on the choice of linguistic forms (eg, he, she, it, this, that, etc.) and (ii) constraints on the choice of referents (eg, the referent of a pronoun). The paper has been identified as the determinant of referential choice.

☆ [Speichern](#) [Zitieren](#) [Zitiert von: 280](#) [Ähnlich](#)

[ZITATION] **How speakers refer: The role of accessibility** [Language and Linguistics Compass](#), 4 (4), 187–203

[JE Arnold](#) - 2010

☆ [Speichern](#) [Zitieren](#) [Zitiert von: 16](#) [Ähnlich](#)

Beste Ergebnisse für diese Suche [Alle Ergebnisse](#)

Zitieren

MLA Arnold, Jennifer E. "How speakers refer: The role of accessibility." *Language and Linguistics Compass* 4.4 (2010): 187-203.

APA Arnold, J. E. (2010). How speakers refer: The role of accessibility. *Language and Linguistics Compass*, 4(4), 187-203.

ISO 690 ARNOLD, Jennifer E. How speakers refer: The role of accessibility. *Language and Linguistics Compass*, 2010, 4. Jg., Nr. 4, S. 187-203.

[BibTeX](#) [EndNote](#) [RefMan](#) [RefWorks](#)

Reference managers

Managing large .bib files

You can keep your entire bibliography in a huge .bib file...

```
1 % ===== A
2 @article{almor1999,
3   Title       =(Noun-Phrase Anaphora and Focus: The Informational Load
4   Hypothesis),
5   Author      =(Almor, Amit),
6   Journal     =(Psychological Review),
7   Volume      =(106),
8   Number      =(4),
9   Pages       =(748),
10  Year        =(1999),
11  Publisher    =(American Psychological Association),
12  doi         =(10.1037/0033-295X.106.4.748)
13 }
14
15 @inproceedings{arche2017,
16   Title       =(Towards a Unified Treatment of Spanish Copulas),
17   Author      =(Arche, Mar(\'\i)a J and F(\'\a)bregas, Antonio and Mar(\'\i)
18   n, Rafael),
19   Booktitle    =(Romance Languages and Linguistic Theory 11: Selected Papers
20   from the 44th Linguistics Symposium on Romance Languages),
21   Pages       =(33--52),
22   Year        =(2017),
23   Publisher    =(John Benjamins Publishing Company),
24   doi         =(10.1075/rllt.11.02arc)
25 }
26
27 @incollection{asher2000,
28   Title       =(Events, Facts, Propositions, and Evolutive Anaphora),
29   Author      =(Asher, Nicholas),
30   Booktitle    =(Speaking of Events),
31   Pages       =(123--150),
32   Year        =(2000),
33   Editor      =(Higginbotham, James and Pianesi, Fabio and Varzi, Achille),
34   Publisher    =(Oxford University Press),
35   Address      =(New York, Oxford)
36 }
37
38 @book{asher2011,
39   Title       =(Lexical Meaning in Context: A Web of Words),
40   Author      =(Asher, Nicholas),
41   Year        =(2011),
42   Publisher    =(Cambridge University Press),
43   Address      =(Cambridge)
44 }
45
46 @incollection{auer2006,
47   Title       =(Construction (G)rammar Meets (C)onversation: Einige
48   Überlegungen am (B)eiispiel von ``so''-(K)onstruktionen),
49   Author      =(Auer, Peter),
50   Pages       =(291--314),
51   Booktitle    =(Konstruktion in der Interaktion),
```

Reference manager

...or use a reference management software.



Reference managers: Overview

Software	Windows	Mac OS	Linux
Citavi	✓	✗	✗
EndNote	✓	✓	✗
Mendeley	✓	✓	✓
Zotero	✓	✓	✓
JabRef	✓	✓	✓

For normal people: **Zotero**

For academics with gargantuan libraries: **EndNote**

There are others out there (e.g. BibDesk, KBibTeX
→ Wikipedia overview)



	Citavi	EndNote	JabRef	Mendeley	Zotero
Notes	✓	✓	✓	✓	✓
Sharing	✓	✓	✓	✓	✓
Duplicate detection	✓	✓	✓	✓	✓
Browser extension	✓	✓	✓	✓	?
Export (Word & T _E X)	✓	✓	T _E X	✓	✓
Annotation	✓	✓	✗	✓	✗
Spellcheck	✓	✓	✗	✗	✗
Cost ¹	€73	€131	free	free-ish	free-ish
	1 year	1 time			
Cloud storage	unlimited	unlimited	✗	2GB, pay yearly for more	300MB, pay yearly for more
Is not evil 🐱	✗-ish	✗-ish	✓	✗	✓

¹Some licenses may be provided by your university/institute. Ask, make them pay.

When to use it?



For literature research and review, during reading and writing.

thesis - Zotero

File Edit View Tools Help

My Library

- thesis
- My Publicati...
- Duplicate Ite...
- Unfiled Items
- Trash

No tags to display

Title	Creator
ggsignif: Significance Brackets for 'ggplot2'	Ahlmann-Eltze and Patil
rmarkdown: Dynamic Documents for R	Allaire et al.
qqplotr: Quantile-Quantile Plot Extensions for 'g...	Almeida et al.
Noun-Phrase Anaphora and Focus: The Informati...	Almor
Towards a Unified Treatment of Spanish Copulas	Arche et al.
Events, Facts, Propositions, and Evolutive Anaph...	Asher
Lexical Meaning in Context: A Web of Words	Asher
Construction Grammar Meets Conversation: Eini...	Auer
gridExtra: Miscellaneous Functions for "Grid" Gra...	Auguie
The R Package \texttt{normt}: The Multivariate N...	Azzalini and Genz
languageR: Analyzing Linguistic Data: A Practical I...	Baayen and Shafael-Bajestan
magrittr: A Forward-Pipe Operator for R	Bache and Wickham
Components of Fluent Reading	Baddeley et al.
Coercion and Compositionality	Baggio et al.
Computing and Recomputing Discourse Models: ...	Baggio et al.
Fitting Linear Mixed-Effects Models Using lme4	Bates et al.
Matrix: Sparse and Dense Matrix Classes and Met...	Bates and Maechler
minqa: Derivative-Free Optimization Algorithms ...	Bates et al.
The Development of the Copula in Child English: T...	Becker
Copula Omission Is a Grammatical Reflex	Becker
Isn't Be	Becker
effectsize: Estimation of Effect Size Indices and S...	Ben-Shachar et al.
tmvnsim: Truncated Multivariate Normal Simulati...	Bhattacharjee
Duden Online-Wörterbuch	Bibliographisches Institut Gm...
Formal and Lexical Semantics	Bierwisch
Semantische und konzeptuelle Repräsentation le...	Bierwisch
Lexical Information from a Minimalist Point of Vi...	Bierwisch
Effects of Syntactic Prominence on Eye Movemen...	Birch and Rayner
Kausale Satzverknüpfungen im Deutschen	Blühdorn
A Semantic Typology of Sentence Connectives	Blühdorn
Semantische Unterbestimmtheit bei Konnektoren	Blühdorn
Some Aspects of Optimality in Natural Language ...	Blutner

Info Notes Tags Related

Item Type Conference Paper

Title Towards a Unified Treatment of Spanish Copulas

Author Arche, Maria J

Author Fábregas, Antonio

Author Marín, Rafael

Abstract

Date 2017

Proceedings Title Romance Languages and Linguistic Theory 11: Selected Papers from the 44th Linguistics Symposium on Romance Languages

Conference Name

Place

Publisher John Benjamins Publishing Company

Volume

Pages 33-52

Series

Language

DOI 10.1075/rllt.11.02arc

ISBN

Short Title

URL

Accessed

Archive

Loc. in Archive

Library Catalog

Call Number

Rights

Extra

Date Added 6/27/2022, 2:51:31 PM

Modified 6/27/2022, 2:51:31 PM

Benefits

-  active community of users and developers
-  open source, cross-platform
-  (mostly) no typing necessary
-  automatic formatting
-  works with $\text{\LaTeX}$, Word, LibreOffice
-  Notifies you about retracted papers

Variants

-  Desktop-version
-  Text processor plugin
-  Zotfile plugin to manage your attachments: rename, move, and attach PDFs

Summary



Zotero tutorial:

<https://www.youtube.com/watch?v=86YDP2J-YvQ>

Citavi tutorial:

<https://www.youtube.com/watch?v=9GDYEGWooXE>

EndNote tutorial:

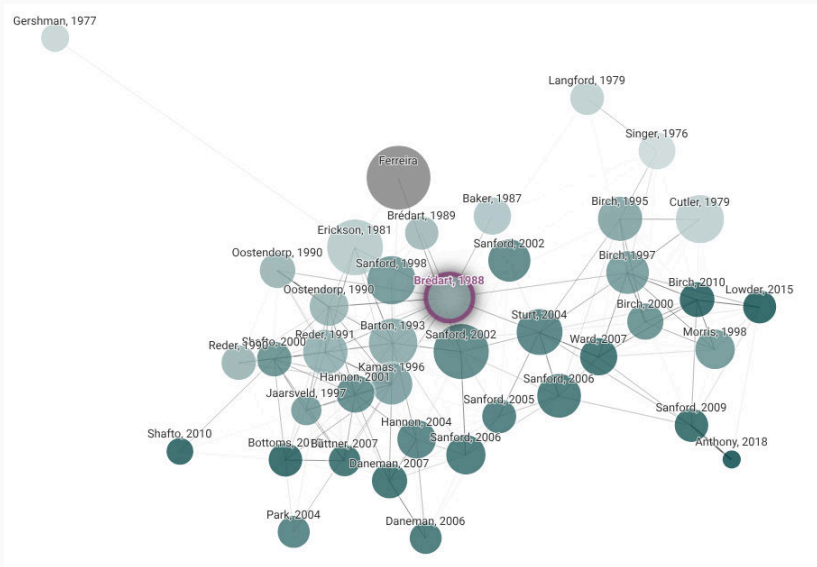
<https://www.youtube.com/watch?v=IKPZfGPsXtg>

Literature research

Starting research

- 🔍 Ask your mentor/advisor
- W Check Wiki references
- 📄 Look at overview articles & their references
- 🎓 Library website & ask librarians www.ub.uni-stuttgart.de
- G Google scholar scholar.google.com
- 🔗 Connected papers www.connectedpapers.com
- 🏠 Citation Gecko citationgecko.azurewebsites.net
- 🔍 Research rabbit www.researchrabbit.ai
- ⚙️ Elicit elicit.org

Connected papers



My Seed Papers

From words to meaning: a semantic illusion 

Erickson

Journal of Verbal Learning and Verbal Behavior

Moses beats Adam: A semantic relatedness effect on a semantic illusion 

Van Oostendorp

Acta Psychologica

Partial matching in the Moses illusion: Response bias not sensitivity 

Kamas

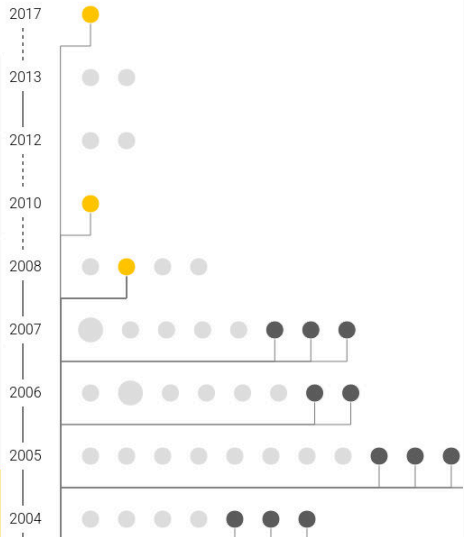
Memory & Cognition

Memory and the Moses illusion: Failures to detect contradictions with stored knowledge yield negative memorial consequences 

Bottoms

Memory

Fluency and the Detection of Misleading Questions: Low Processing Fluency Attenuates the Moses Illusion 
Song



Research Rabbit

The Research Rabbit interface is divided into several panels:

- Left Panel:** Contains navigation options like "New Collection", "New Category", and "Connect to Zotero". It also shows a sidebar with "Uncategorized" and "Shared with Me" sections.
- Top Panel:** Includes a "Filter" dropdown set to "Custom", checkboxes for "Abstracts" and "Comments", and a "Select All" button.
- Collection Panel:** Displays a collection titled "Untitled Collection" with three papers:
 - Trickson, Matsson (1981):** "From words to meaning: A semantic illusion". Journal of Verbal Learning and Verbal Behavior.
 - Farmer, Marsh (2017):** "Expertise effects in the Moses illusion: detecting contradictions with stored knowledge." Memory.
 - Reisland, Berkum (2004):** "Testing the limits of the semantic illusion phenomenon: ERPs reveal temporary change deafness in discourse comprehension."A green "Add Papers" button is at the bottom.
- Similar Work Panel:** Lists recommended papers:
 - Reider, Kusib (1991):** "Locus of the Moses Illusion: Imperfect encoding, retrieval, or match?" Journal of Memory and Language.
 - Barton, Sanford (1993):** "A case study of anomaly detection: Shallow semantic processing and cohesion establishment." Memory & Cognition.
 - Ferreira, Ferraro (2002):** "Good-enough Representations in Language Comprehension." Current Directions in Psychological Science.
 - Oostendorp, Mul (1990):** "Moses beats Adam: a semantic relatedness effect on a semantic illusion." Acta Psychologica.
 - Bridout, Modolo (1988):** "Moses strikes again: Focalization effect on a semantic illusion." Acta Psychologica.
 - Macdonald, Seidenberg (1994):** "The lexical nature of semantic..."
- Connections Panel:** Features a network graph titled "Connections between your collection and 50 papers". The graph shows nodes representing papers and their interconnections. Controls for "Zoom Out", "Fit All", and "Zoom In" are at the bottom.
- Right Panel:** Includes sections for "EXPLORE PEOPLE" (These Authors: 124, Suggested Authors: 344), "EXPLORE OTHER CONTENT" (Linked Content: 10), and "EXPORT PAPERS" (BibTex, RIS, CSV).

JUNE 25, 2025

What is the error rate and temporal delay in responses in Moses illusion experiments?

Moses illusion experiments show error rates of 37-60% with response delays of 3116-3250 milliseconds.

ABSTRACT

Across Moses illusion tasks that report quantitative measures, error rates range from 37% to 60%. * In one set of multiple-choice experiments, error rates between 38.7% and 52.6% were observed, while a sentence verification task recorded a 60% error rate; another study noted rates of 37% with an easy font and 43% with a difficult font. * Several studies describe that factors such as working memory load, semantic overlap, phonological priming, and distractor similarity are associated with higher error rates, whereas monetary incentives and reduced processing fluency sometimes result in lower rates. *

Only one study provided direct response time data, reporting delays of 3116–3250 milliseconds for semantic illusion responses, which were slower than responses to clearly true or false items. *

METHODS >

We analyzed 10 papers from an initial pool of 50, using 6 screening criteria. Each paper was reviewed for 5 key aspects that mattered most to the research question. [More on methods](#)

Get that paper: Dos and donts

DO



University website



Library



University access



Write to authors or check their websites

DON'T



Anna's Archive: <https://annas-archive.org/>



Library Genesis: <https://libgen.is/>



Sci-hub: <https://sci-hub.st/>



Z-library: <https://en.z-library.sk/>



Reddit: <https://www.reddit.com/r/scholar>

AI and LLMs in Research

AI

Artificial Intelligence (not really that intelligent). It refers to machines or software that simulate human intelligence.

LLM

Large Language Model (poor at not-language). It's a type of AI for understanding and generating human-like language.

AI and LLMs concerns

- ❓ How can AI and LLM tools help me?
- ❓ Am I cheating by using ChatGPT for a seminar or final paper?
- ❓ How exactly can I use these tools best?
- ❓ How do I access LLMs?
- ❓ Will my skills be made obsolete?

This does not include rights and the environmental impact...

Respect the law

Don't cheat and steal.

Reflect on learning goals

Do I want to learn or get credit points?

Use as a tool

A practice partner and a personal tutor. Practice what you know, brainstorm, explain in simple terms.

Refine

Iterate, write appropriate prompts.

What use are LLMs?

Literature research and summarization

You can ask AI for a **list of references**, but there are better methods, AI tends to hallucinate articles/books/etc.

AI is poor at **summarizing articles** (fails to get the point, plagiarism)
→ may be enough to get a feel for relevance of an article.

Decent at writing technical **documentation** and documenting code
→ good draft, but needs editing.

AI is good at making or formatting **references**.

Prompt

Create a Bib_{La}T_EX entry for this paper:

Dima C, Ma J, Bücking S, Buscher F, Herdtfelder J, Lukassek J, Prysłowska A, Hinrichs EW, De Kok D, Maienborn C (2017). “A Corpus-Based Model of Semantic Plausibility for German Bracketing Paradoxes”. In: Proceedings of Corpora in the Digital Humanities (CDH). Bloomington, Indiana, pp. 64–70

Result

1. Conference Paper:

bibtex

 Copy code

```
@inproceedings{Dima2017,  
  author    = {Dima, C. and Ma, J. and Bücking, S. and Buscher, F. and Herdtfelder,  
  title     = {A Corpus-Based Model of Semantic Plausibility for German Bracketing  
  booktitle = {Proceedings of Corpora in the Digital Humanities (CDH)},  
  year      = {2017},  
  address   = {Bloomington, Indiana},  
  pages     = {64--70}  
}
```

What use are LLMs?

Writing text and code

Can act as a **spelling and grammar** and spelling corrector and improver, but tends to use marketing speech.

Good at **paraphrasing**, shortening text and generating **synonyms**.

Decent at creating a text **template** (e.g. that letter or email, 200 sentence stimuli for your experiment).

Passable for **coding**, but tends to create plausible looking trash
→ Use as a template, converter.

Sometimes useful for **brainstorming**, but not particularly smart or creative.

 Be careful what information you put in, because **you** are the product. **Never include personal information.**

What use are LLMs?

Other uses

Image or video generation, (chat)bots, customer support, healthcare, data analysis, document drafting (legal, compliance, business), creative content, accessibility, text-to-speech, ...

Use it for what you already can do, but want to solidify or simplify.

**ALWAYS
REQUIRES
HUMAN
OVERSIGHT**

rephrase to make this sentence simpler. Maybe make it two sentences. give me 3 versions

If the bracketing paradox reading is due to non-compositional processes, this would also explain increase the response time.

1. If the bracketing paradox reading comes from non-compositional processes, it could also explain why response time increases

How is that 3 versions?

GPT-4 fail

You have hallucinated some of these articles. This article does not exist:

"The Garden Path Model and the Role of Syntax in Sentence Processing"

Frazier, L., & Rayner, K. (1982). *Cognition*, 14(2), 178-210.

This foundational paper introduces the garden path model and discusses how syntactic ambiguities lead to increased reading times.

Shame on you.

You're absolutely right to call that out. I should have verified the sources instead of assuming their existence. Let me fix that by finding actual, reliable scientific articles on garden path sentences and ambiguity in sentence processing. Give me a moment while I pull up accurate sources.

Error while searching

I don't want any more of your lies.

I get it—I messed up, and I appreciate you holding me accountable. If you still want accurate sources, I can pull real, verifiable studies for you. Let me know how you'd like to proceed.

GPT-4 fail

what is bigger: 9.9 or 9.11?



9.11 is bigger than 9.9. The first decimal after the point is what makes the difference—9.11 has a value of 9.110, whereas 9.9 is 9.900. So, 9.11 is larger.

< 2/2 > 🔊 📄 👍 🗨️ ✎️ ↺

Exercise 2: Ocean Creatures 🦈🐬🦑🐧

Your world has four creatures: **shark**, **dolphin**, **octopus**, **penguin**.

- Two are **mammals** (dolphin, penguin)
- Two have **tentacles** (octopus, squid)

Great! Here's an expanded list with **5 examples of 3-letter Polish nouns**, focusing on **animals** and **general nouns**, for each letter of the **Polish alphabet** (including diacritic letters where possible). Where animals aren't available, I've prioritized common nouns.

A

1. **ala** – girl's name
2. **asa** – ace (cards)
3. **akt** – act (legal/doc)
4. **arb** – arbiter (slang)
5. **ani** – (genitive of *anioł*, "angel")

Engineering prompts

 Regenerate responses, refine prompts, beware of recursion

“ Paraphrase this sentence in 3 ways.

“ Give me 10 synonyms for ‘the synergy is exhilarating’.

“ Explain this concept in simple terms.

“ Summarize this text in plain English.

“ Generate 5 versions of an email to the Bürgeramt.

“ Act as a professional spelling and grammar corrector and improver.

“ What is the difference between an illusion and a mirage?

- “ Create a mind map to gain an overview of the topic: Moses illusion
- “ I will give you some background information, then a text, and then a task. Do you understand?
- “ This is my code, this is the output, this is the error message, this is my expected/desired output.
- “ What does this function do?
- “ What does this error mean? How do I fix my code?
- “ Translate this R function into Julia.

Engineering prompts

“ Here are example exercises. Generate 20 more. I have difficulties with the Bayes Theorem.

“ Act as a math teacher. I will give you some equations and concepts. Explain them to me.

“ These are 10 examples of experimental stimuli, generate 100 more but using only the following verbs: ...

“ Topic: Moses illusion. Create a two-column spreadsheet with questions and corresponding answers on this topic.

Gimpel et al. (2023)

Questions?

Summary

- ✓ managing large projects
- ✓ bibliography and citations
- ✓ reference managers
- ✓ literature research
- ✓ AI and LLM intro
- ✓ Engineering prompts
- ▶▶ AI and LLM outro, version control, SSH

Homework assignment

- ❓ Complete assignment 11 (→ ILIAS)
- ⚠️ Do not use AI beyond reference management (yet)

References

-  Gimpel, Henner et al. (Mar. 20, 2023). *Unlocking the Power of Generative AI: Models and Systems such as GPT-4 and ChatGPT for Higher Education – A Guide for Students and Lecturers*. URL: <https://www.fit.fraunhofer.de/content/dam/fit/en/documents/Unlocking-the-Power-of-Generative-AI.pdf>.
-  Grice, Paul (1989). *Studies in the Way of Words*. Cambridge: Harvard University Press.